

Mini Assessment

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Q1. If an API returns 429, what should your program do?

Answer: A 429 status response indicates 'Too Many Requests', meaning the application has hit the rate limit threshold. The client program must halt active requests and handle it systematically:

- Check for a 'Retry-After' header to respect the exact wait period specified by the server.
- Introduce an exponential backoff routine to increase delay spacing dynamically between subsequent retries.
- Implement request throttling to prevent the client application from entering a continuous retry loop.
- Gracefully catch the failure state and log appropriate diagnostic details without causing system crashes.

Q2. If response.json() fails, what should your program do?

Answer: A failure in response.json() typically triggers a parsing exception (such as JSONDecodeError). This occurs when the API returns raw text, HTML error blocks, or empty strings. The routine should handle it as follows:

- Enclose the parsing statement within a robust try-except structural block to isolate decoding exceptions.
- Inspect response.status_code prior to decoding to ensure a valid payload exists.
- Capture and log response.text directly to capture the exact payload received from the endpoint.
- Provide a safe fallback data structure (like an empty dictionary) to allow continuing execution safely.

Q3. If a JSON key is missing, why might .get() be safer than []?

Answer: Utilizing square brackets like data['key'] will throw a hard KeyError if the key is absent, terminating application execution. Adopting the .get() method offers significant advantages:

- It safely returns a default fallback state (None) instead of breaking execution routines.
- It supports custom configurations by allowing a custom backup value, e.g., data.get('key', 'N/A').
- It simplifies source code readability by reducing the necessity for extensive conditional if-else key verification steps.
- It builds resilient production software capable of tolerating minor inconsistencies in external data schema.

End of Mini Assessment